

Order of Operations: Library Scenario Setup

An AccessControlSystem is created to act as the receiver, along with a CampusComponent representing the library area. A SecureAreaCommand is then created, linking the access control system to that area.

A CommunicationService is created for the library staff, acting as a FacilityStaff. Its state is changed once, moving it into Detection. At this point the staff does not yet have a command assigned to it.

A second CommunicationService is created for library security, acting as a CampusSecurity. Its state is also changed once, placing it into Detection.

An IssueAlertCommand is created, linking the security receiver to the issue message "Student strike".

An EmergencyCoordinator is created to mediate between staff and security. The staff's mediator is set to this coordinator, and security is registered as a component the staff can notify.

A second EmergencyCoordinator is created for security's side. Security's mediator is set to this new coordinator, and both the staff and the access control system are registered as components security can notify.

The staff's state is changed again, moving from Detection to Response, and the staff's mediator then notifies the relevant components of this change.

Security's state is changed from Detection to Response, and security's mediator notifies its registered components in turn.

The staff's state changes once more, continuing its progression through the sequence.

The operator console executes the security alert command, followed by the library security management command.

The staff's state is changed again, representing the point where security begins resolving the issue.

The operator console undoes the last two executed commands, in reverse order.

Finally, all heap-allocated objects created directly by the invoker and the mediators are deleted: the operator console and both mediators.